QUICK DIAGNOSTIC CHART [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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X: Applied

Troubleshooting item		Possible factor: Starter motor malfunction (mechanical or electrical)	Starter circuit including ignition switch is open	Improper engine oil level	Low or dead battery	Charging system malfunction	Improper engine compression	Improper oil level gauge	Base engine malfunction	Drive plate (ATX) or flywheel (MTX) is seized	Improper tension or damaged drive belts	Improper engine coolant level	Water and anti-freeze mixture is improper	Cooling system malfunction (such as radiator, hoses,overflow system, thermostat)	Cooling fan system malfunction	Insufficient battery recharge
1	i-stop warning light (amber) illuminates					X										
2	i-stop warning light (amber) flashes	X	X		X	X										
3	i-stop indicator light (green) flashes															
4	i-stop function does not operate	X	X		X	X										
5	i-stop function operates under no engine-stop conditions															
6	Engine does not restart	X	X		X	X										
7	Limited i-stop function operation time				X	X										X
8	Ineffective operation of A/C during engine stop (i-stop system)															
9	Acceleration malfunction (ATX)															
10	Vehicle moves when accelerating from standstill on slope (ATX)															

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X: Applied

Troubleshooting item		Possible factor: PCM malfunction	Engine or transaxle mounts are improperly installed	Cooling fan or condenser fan seats improper	Fuel quality	Improper engine oil viscosity	Improper valve timing	Oil leakage	Improper oil or fluid level	Coolant leakage	Electric variable valve timing system malfunction	Hydraulic variable valve timing system malfunction	Engine overheating	Air cleaner element clogged or restricted	Air leakage from intake-air system (loose tubes, cracks, gaskets breakage)	Air suction in intake-air system	Intake-air system restriction
1	i-stop warning light (amber) illuminates						X				X	X				X	
2	i-stop warning light (amber) flashes	X															
3	i-stop indicator light (green) flashes																
4	i-stop function does not operate																
5	i-stop function operates under no engine-stop conditions																
6	Engine does not restart															X	
7	Limited i-stop function operation time																
8	Ineffective operation of A/C during engine stop (i-stop system)																
9	Acceleration malfunction (ATX)																
10	Vehicle moves when accelerating from standstill on slope (ATX)																

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	Possible factor / Troubleshooting item	Drive-by-wire control system improper operation	Drive-by-wire control system operates with brake override system	Throttle body malfunction	Purge solenoid valve malfunction	Vacuum leakage (vacuum hose damage, misrouting)	Ignition coil malfunction (e.g. open, short or cracks)	Initial ignition timing misadjustment (CKP and pulse wheel misadjustment)	Spark plug malfunction	Fuel pump (low-pressure side) malfunction (mechanically or electrically)	Pressure regulator (built-into fuel pump unit at fuel tank) malfunction	Inadequate fuel pressure (high-pressure side)	Inadequate fuel pressure (low-pressure side)	Fuel line restricted or clogged
1	i-stop warning light (amber) illuminates	X												
2	i-stop warning light (amber) flashes													
3	i-stop indicator light (green) flashes													
4	i-stop function does not operate													
5	i-stop function operates under no engine-stop conditions													
6	Engine does not restart	X												
7	Limited i-stop function operation time													
8	Ineffective operation of A/C during engine stop (i-stop system)													
9	Acceleration malfunction (ATX)													
10	Vehicle moves when accelerating from standstill on slope (ATX)													

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	Possible factor / Troubleshooting item	Fuel injector malfunction (inoperative)	Fuel filler cap malfunction	Fuel filters restricted or clogged	Relief valve malfunction (built-into high pressure fuel pump)	Spill valve control solenoid valve malfunction (built-into high pressure fuel pump)	Fuel leakage at fuel line	High pressure fuel pump malfunction	Incorrect fuel injection timing or amount	Improper air/fuel ratio mixture control	Exhaust system or catalytic converter restriction	Exhaust gas leakage	Catalytic converter malfunction	PCV valve malfunction	Fuel tank vent system malfunction	Charcoal canister malfunction
1	i-stop warning light (amber) illuminates															
2	i-stop warning light (amber) flashes															
3	i-stop indicator light (green) flashes															
4	i-stop function does not operate															
5	i-stop function operates under no engine-stop conditions															
6	Engine does not restart															
7	Limited i-stop function operation time															
8	Ineffective operation of A/C during engine stop (i-stop system)															
9	Acceleration malfunction (ATX)															
10	Vehicle moves when accelerating from standstill on slope (ATX)															

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	Possible factor / Troubleshooting item	Constant voltage supply circuit malfunction	Main relay malfunction (mechanically or electrically)	No battery power supply to PCM	Poor PCM ground or vehicle ground (open)	Electrical connector disconnection	Oil pressure switch or related circuit malfunction	ECT sensor or related circuit malfunction	Transaxle range sensor malfunction (ATX)	Brake switch or related circuit malfunction	A/F sensor or HO2S or related circuit malfunction	IAT sensor No.1 (integrated in MAF sensor/ IAT sensor No.1) or related circuit malfunction	IAT sensor No.2 (integrated in MAP sensor/ IAT sensor No.2) or related circuit malfunction	Fuel pressure sensor or related circuit malfunction	MAP sensor or related circuit malfunction	Starter interlock switch malfunction (MTX)	Starter interlock switch or related circuit malfunction (MTX)
1	i-stop warning light (amber) illuminates																
2	i-stop warning light (amber) flashes								X								
3	i-stop indicator light (green) flashes								X								
4	i-stop function does not operate							X	X	X		X					
5	i-stop function operates under no engine-stop conditions								X								
6	Engine does not restart							X	X	X							
7	Limited i-stop function operation time																
8	Ineffective operation of A/C during engine stop (i-stop system)																
9	Acceleration malfunction (ATX)					X			X								
10	Vehicle moves when accelerating from standstill on slope (ATX)																

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Troubleshooting item		Possible factor: Intake CMP sensor damaged (e.g.open or short circuit)	Intake CMP sensor pulse wheel (intake camshaft) damaged	Exhaust CMP sensor damaged (e.g.open or short circuit)	Exhaust CMP sensor pulse wheel (exhaust camshaft) damaged	Improper gap between CKP sensor and pulse wheel	CKP sensor damaged (e.g.open or short circuit)	CKP sensor pulse wheel damaged	CPP switch or related circuit malfunction (MTX)	Neutral switch or related circuit malfunction (MTX)	MAF sensor or related circuit malfunction	KS or related circuit malfunction	TP sensor or related circuit malfunction	APP sensor or related circuit malfunction	Improper vehicle speed signal	Cruise control system operation improperly
1	i-stop warning light (amber) illuminates						X	X								
2	i-stop warning light (amber) flashes															
3	i-stop indicator light (green) flashes									X						
4	i-stop function does not operate								X	X				X	X	
5	i-stop function operates under no engine-stop conditions									X				X		
6	Engine does not restart								X	X				X	X	
7	Limited i-stop function operation time													X		
8	Ineffective operation of A/C during engine stop (i-stop system)															
9	Acceleration malfunction (ATX)															
10	Vehicle moves when accelerating from standstill on slope (ATX)															

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	Possible factor / Troubleshooting item	A/C system improper operation	Improper refrigerant charging amount	A/C relay (A/C control signal circuit) malfunction	A/C compressor magnetic clutch malfunction	Clutch slippage (MTX)	ATX related parts malfunction (ATX)	Improper ATF level (ATX)	Brake dragging	Loose parts	Improper balance of wheels and tires	Drive line malfunction	Suspension malfunction	Immobilizer system operating properly	Immobilizer system or related circuit malfunction	Advanced keyless entry system malfunction	i-stop system improper operation	i-stop related part malfunction	DC-DC converter malfunction	DC-DC converter system error
1	i-stop warning light (amber) illuminates																X			
2	i-stop warning light (amber) flashes						X												X	
3	i-stop indicator light (green) flashes																X			
4	i-stop function does not operate	X					X											X		
5	i-stop function operates under no engine-stop conditions	X																		
6	Engine does not restart	X					X											X		
7	Limited i-stop function operation time																			
8	Ineffective operation of A/C during engine stop (i-stop system)	X																		
9	Acceleration malfunction (ATX)						X													
10	Vehicle moves when accelerating from standstill on slope (ATX)						X													X

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X: Applied

	Possible factor / Troubleshooting item	i-stop OFF switch malfunction	i-stop OFF signal malfunction	Instrument cluster malfunction	Driver-side buckle switch malfunction	Door latch switch or trunk lid latch switch (4SD)/ liftgate latch switch (5HB) malfunction	Bonnet latch switch malfunction	Battery malfunction	Ambient temperature sensor malfunction	Current sensor malfunction	ABS wheel-speed sensor malfunction	Clutch stroke sensor malfunction (MTX)	Steering angle sensor malfunction	Driver-side air mix actuator malfunction	Driver-side air mix actuator position sensor malfunction	Climate control unit malfunction	Battery fluid temperature sensor malfunction	Keyless antenna malfunction
1	i-stop warning light (amber) illuminates	X	X	X	X	X	X											
2	i-stop warning light (amber) flashes							X		X								
3	i-stop indicator light (green) flashes					X												
4	i-stop function does not operate	X			X	X	X		X		X	X	X	X		X	X	X
5	i-stop function operates under no engine-stop conditions	X		X	X	X	X		X		X	X	X	X	X	X		
6	Engine does not restart				X	X					X	X	X					
7	Limited i-stop function operation time	X						X				X	X	X		X		
8	Ineffective operation of A/C during engine stop (i-stop system)								X					X	X			
9	Acceleration malfunction (ATX)																	
10	Vehicle moves when accelerating from standstill on slope (ATX)																	

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	Possible factor / Troubleshooting item	Heater core temperature sensor malfunction	Power brake unit malfunction	Cabin temperature sensor malfunction	Solar radiation sensor malfunction	Number of times i-stop function operates exceeds assured operation amount of components	Brake fluid pressure sensor malfunction	M position switch malfunction (ATX)	Low-G (XY) sensor malfunction	TFT sensor malfunction (ATX)	Electric AT oil pump malfunction (ATX)	DSC HU/CM malfunction	Generator drive belt wear	Generator malfunction	SAS control module unit malfunction	EPS control module malfunction	ATX malfunction
1	i-stop warning light (amber) illuminates												X				
2	i-stop warning light (amber) flashes					X	X		X			X	X	X	X		
3	i-stop indicator light (green) flashes						X	X				X					
4	i-stop function does not operate		X			X	X	X	X	X							
5	i-stop function operates under no engine-stop conditions		X	X		X		X	X							X	
6	Engine does not restart		X				X	X		X		X	X				
7	Limited i-stop function operation time	X	X				X	X				X	X	X		X	
8	Ineffective operation of A/C during engine stop (i-stop system)	X		X	X												
9	Acceleration malfunction (ATX)						X		X		X	X					X
10	Vehicle moves when accelerating from standstill on slope (ATX)						X		X		X	X					X

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